

Meeting 05.08.2026

Weekly update with Nils

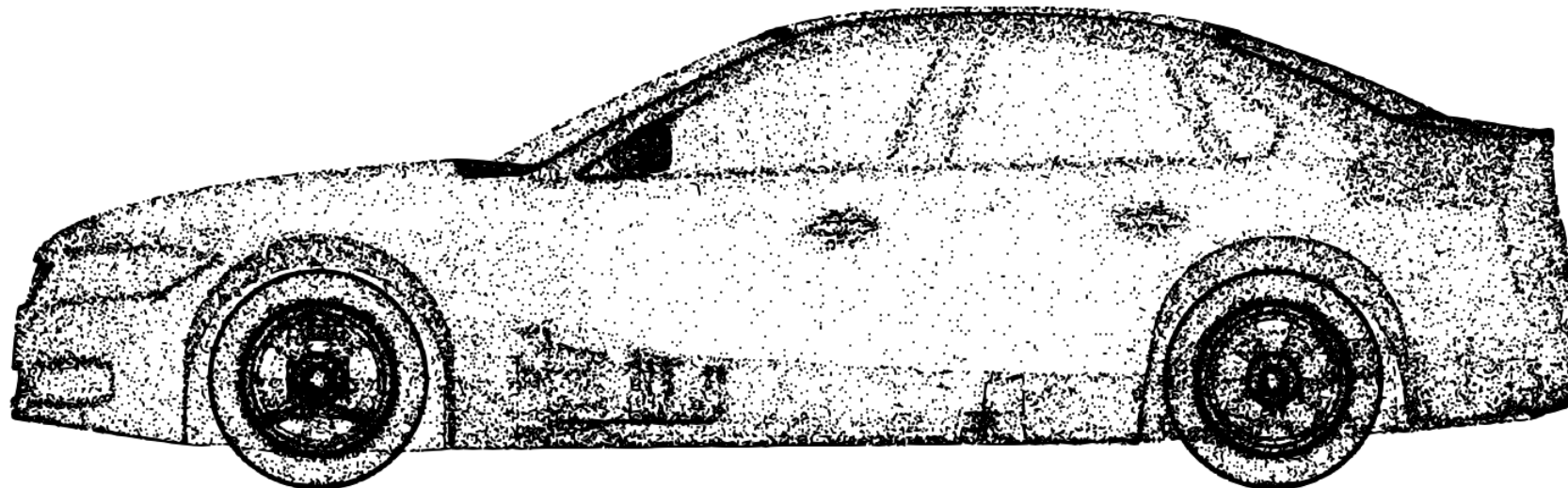
Experiment Design

X-Axis Aligned Sine Shift – 100% intensity

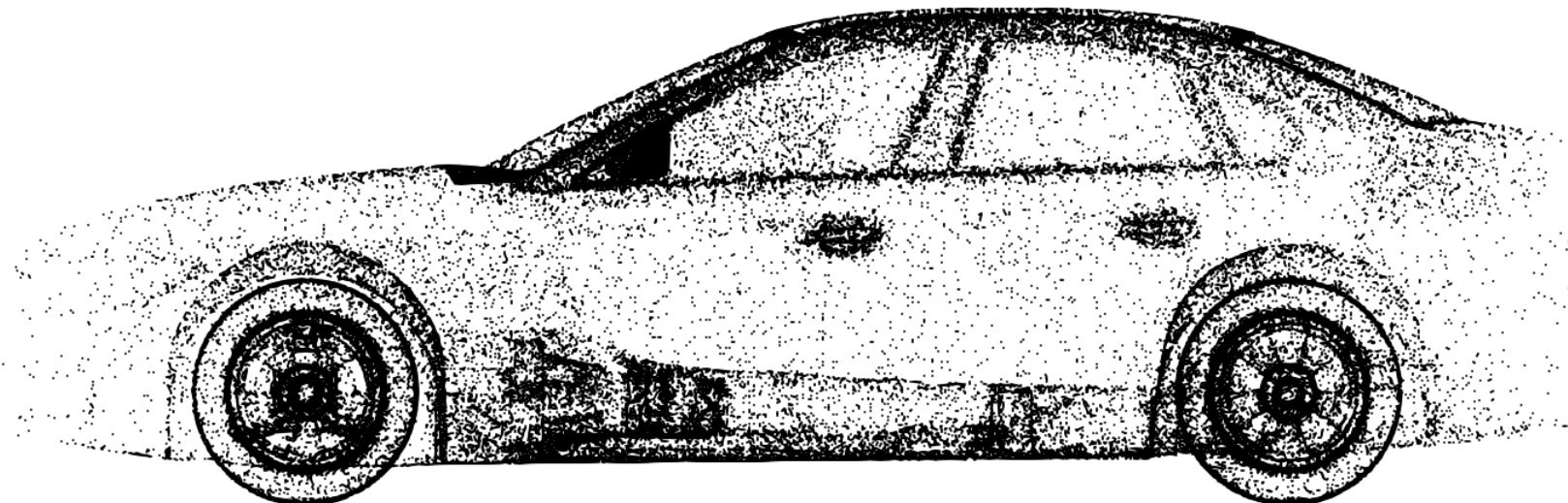


X-Axis

Before



After



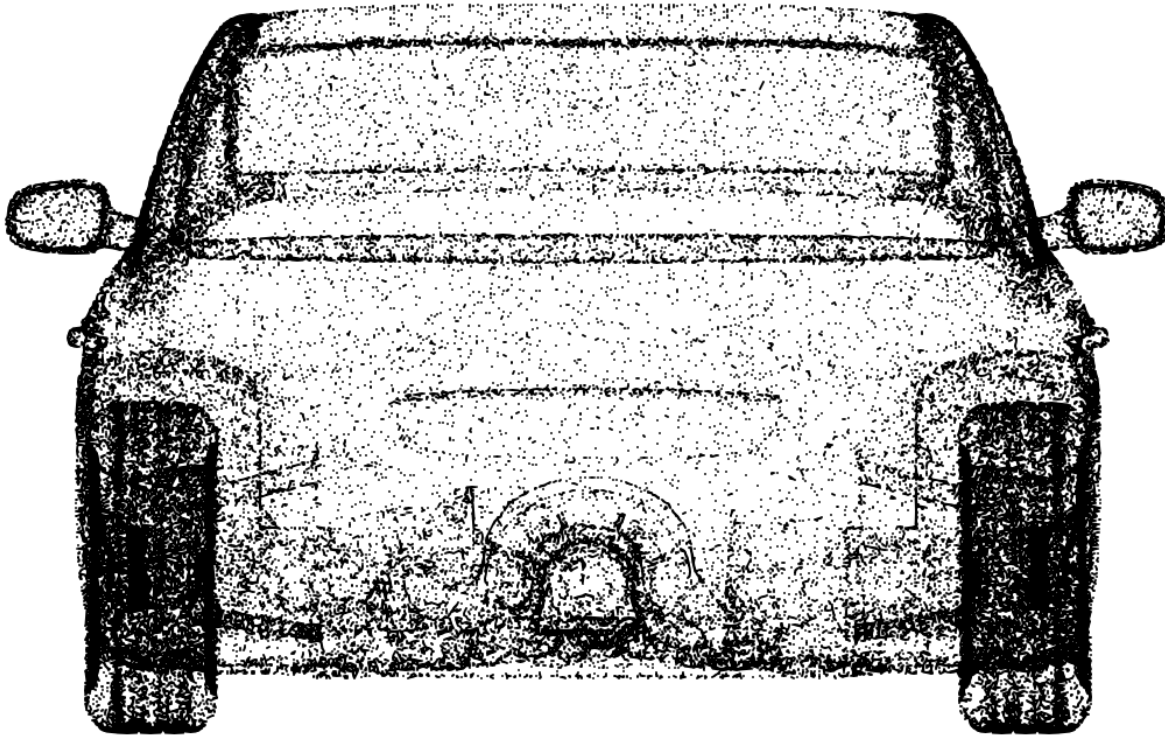
Both are 131k

Y-Axis Aligned Sine Shift – 100% intensity

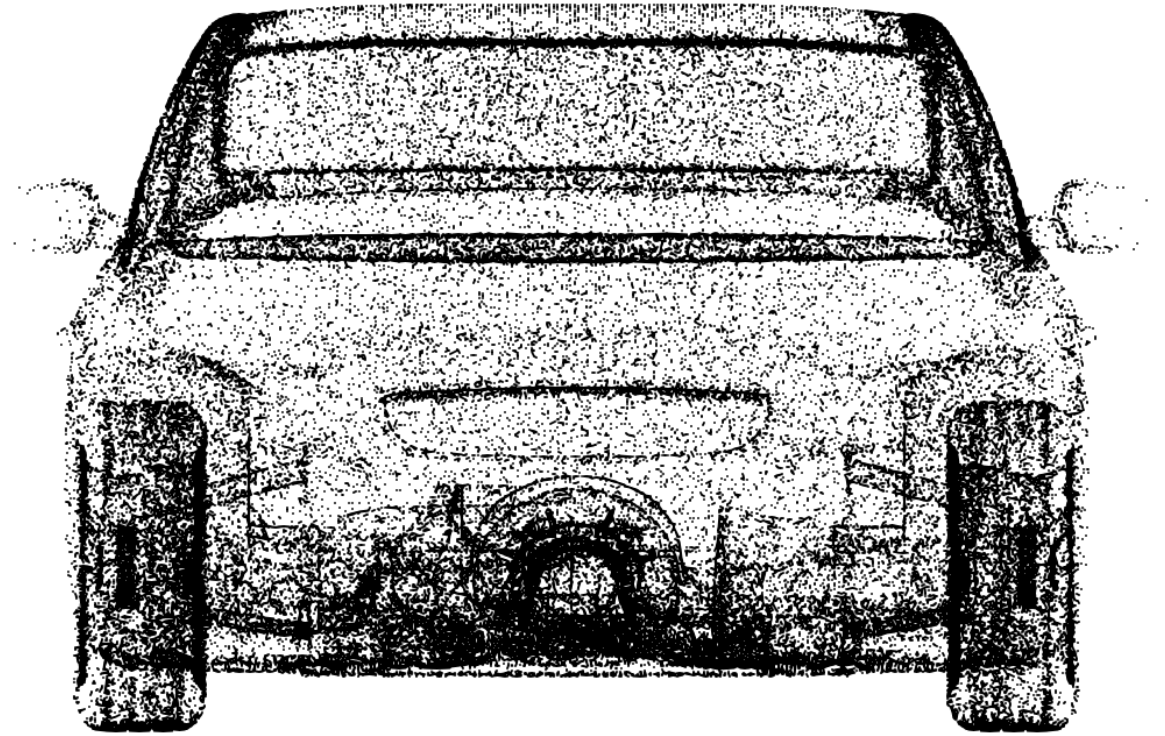


Y-Axis

Both are 131k

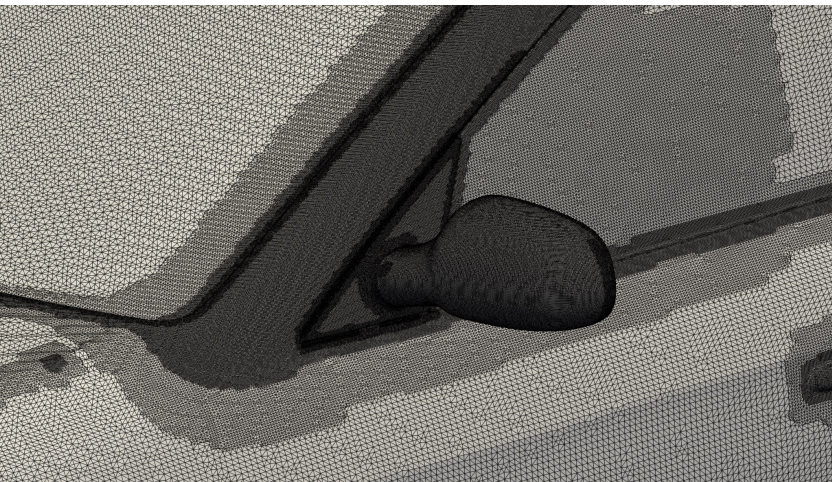


Before



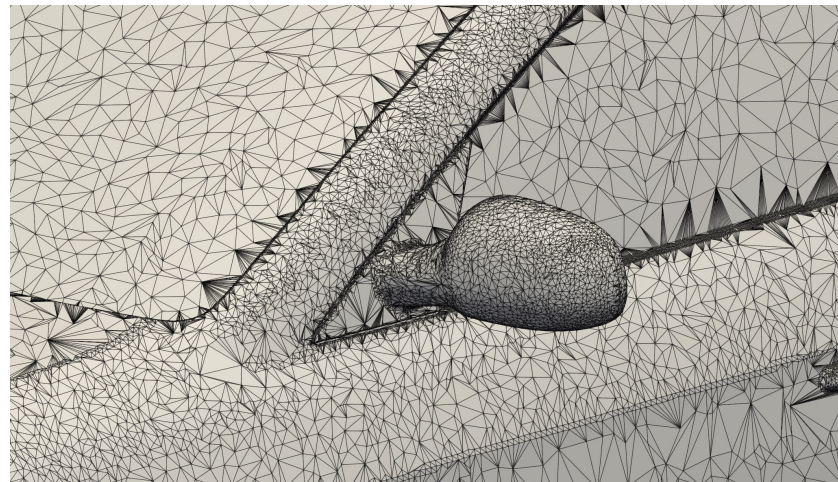
After

Remeshing – Feature based decimation (Fast)



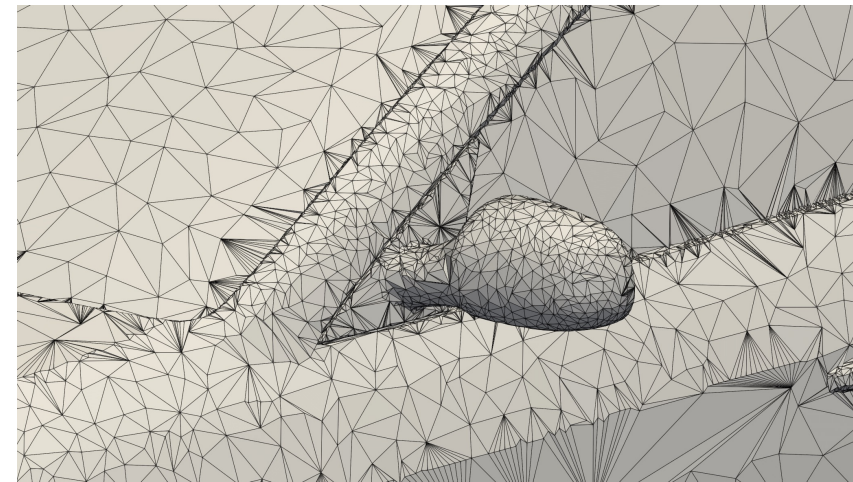
8M Points

Divide by 10



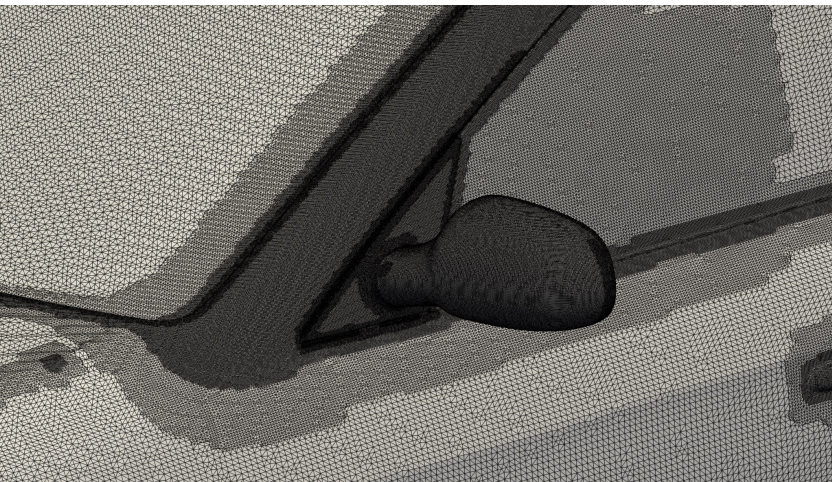
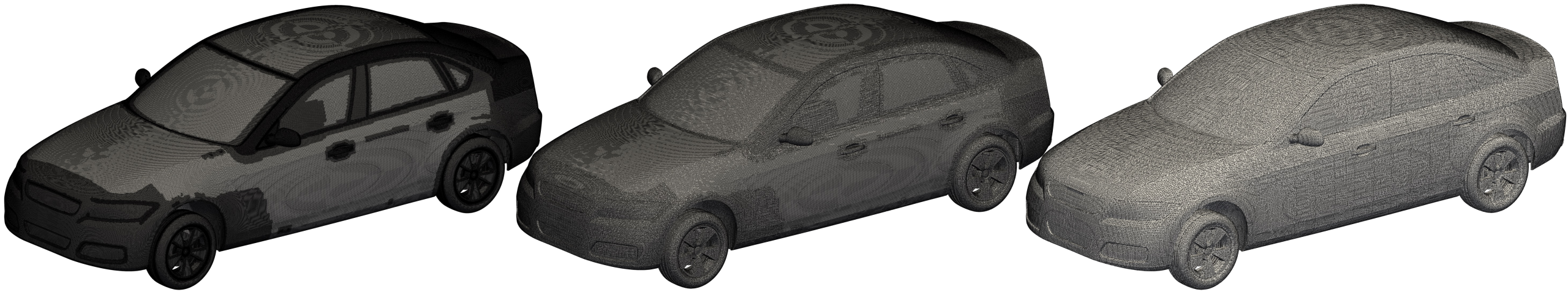
800k Points

Divide by 40

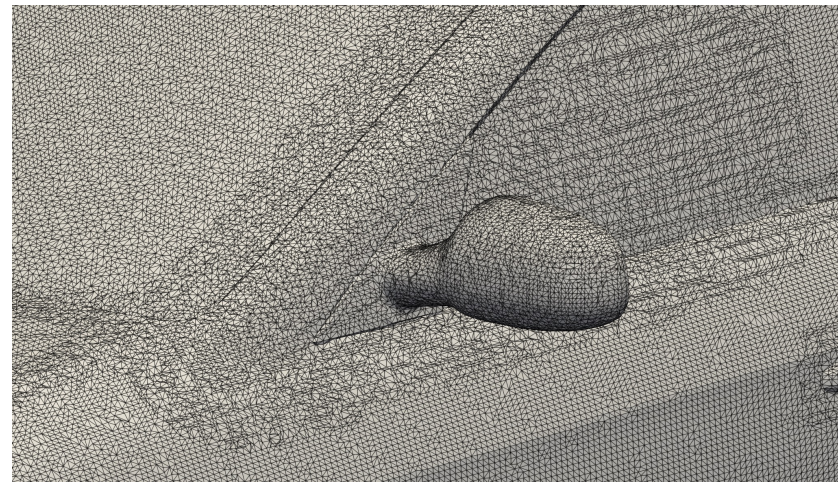


200k Points

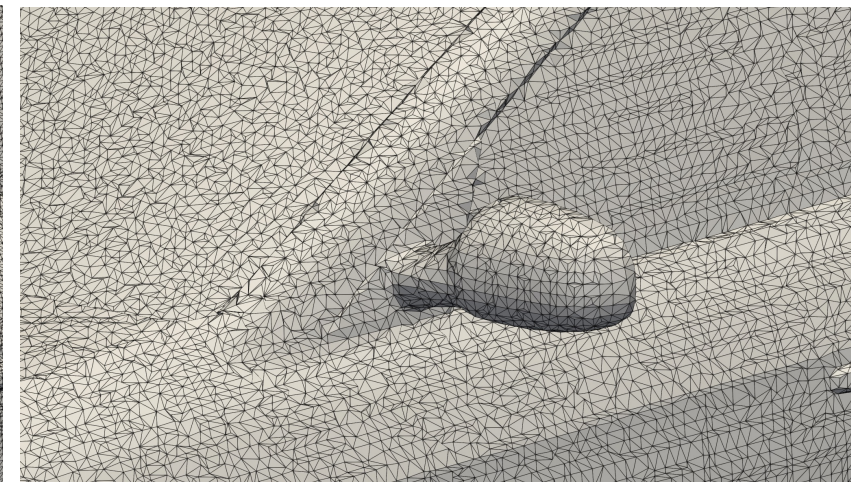
Remeshing – Voxel based decimation (Moderate)



8M Points



800k Points

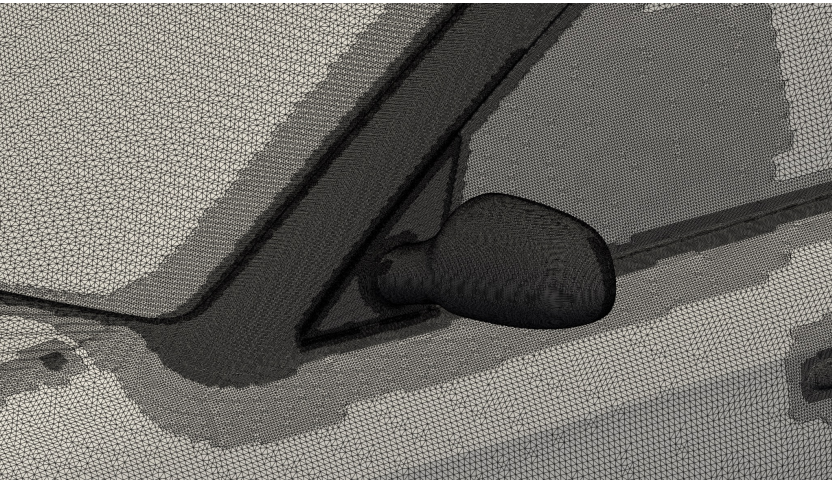
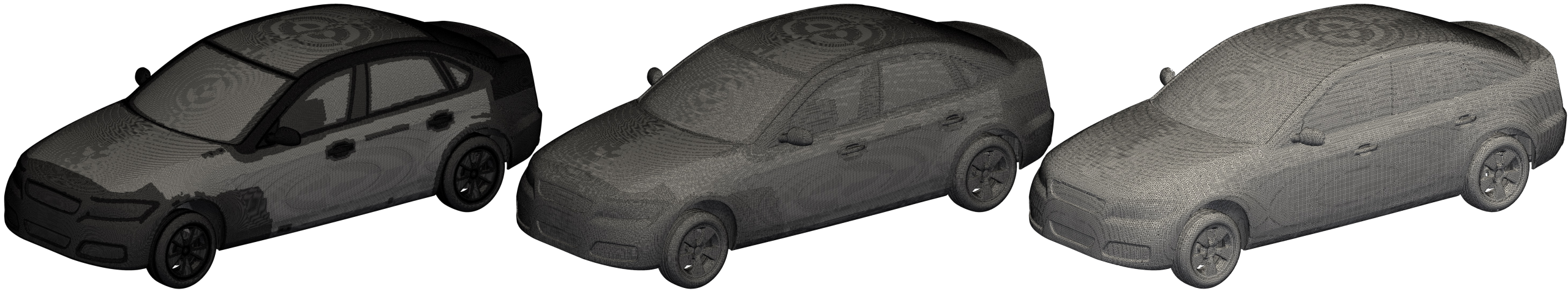


200k Points

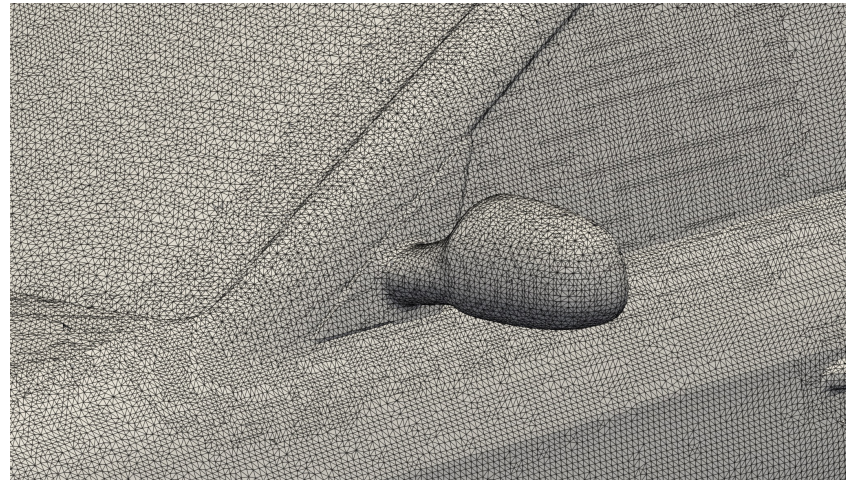
Divide by 10

Divide by 40

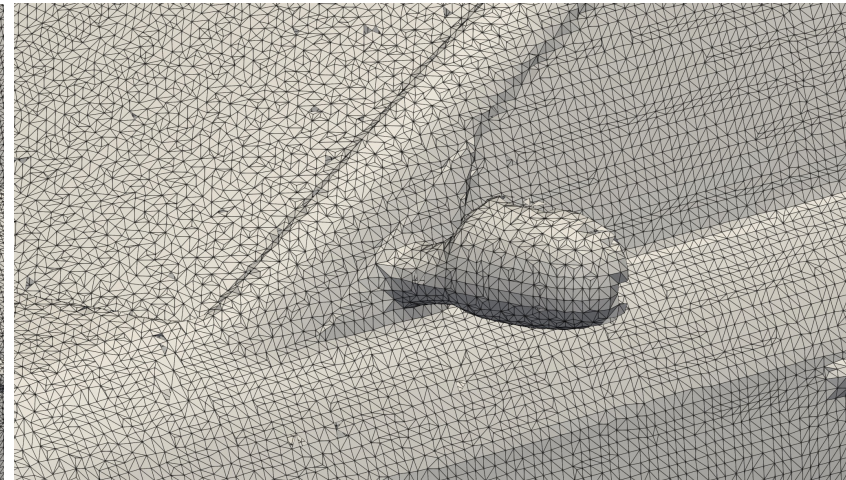
Remeshing – Isotropic decimation (Slow)



8M Points



800k Points



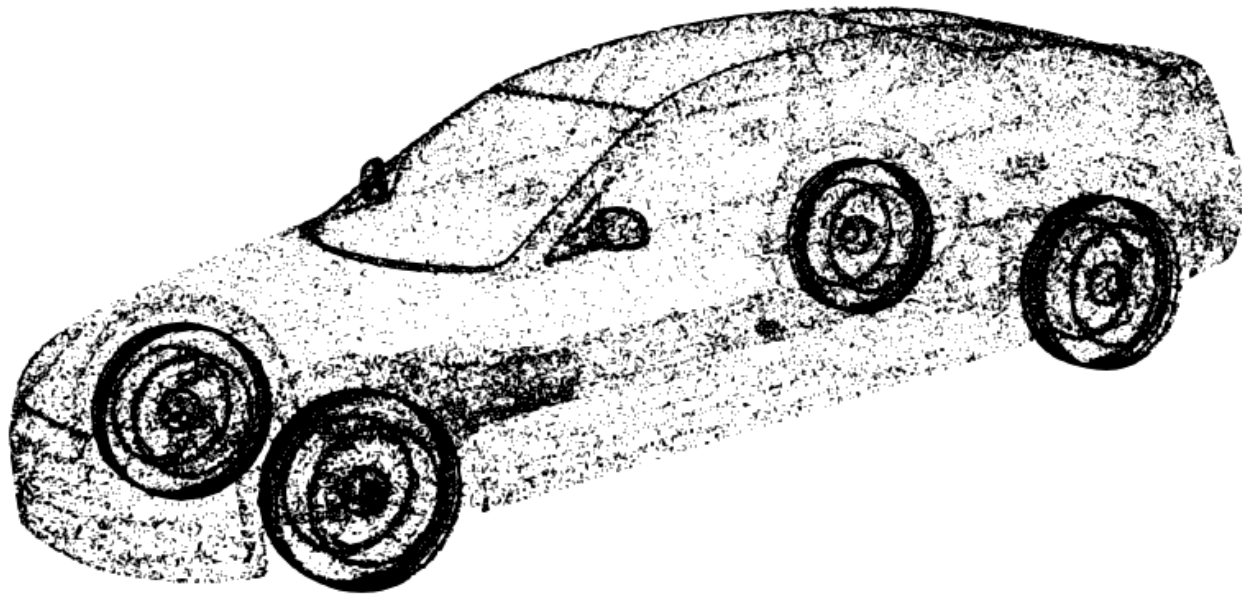
200k Points

Divide by 10

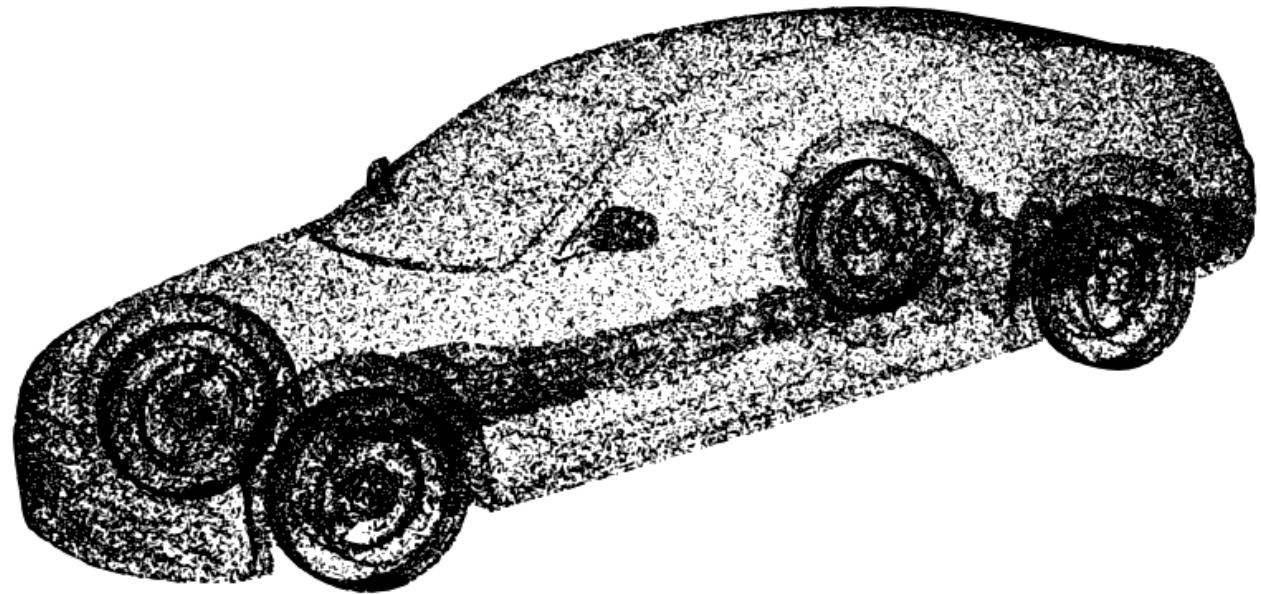
Divide by 40

Results - Ablation

Ablation on different shift range during training based on density shift strength (beta)



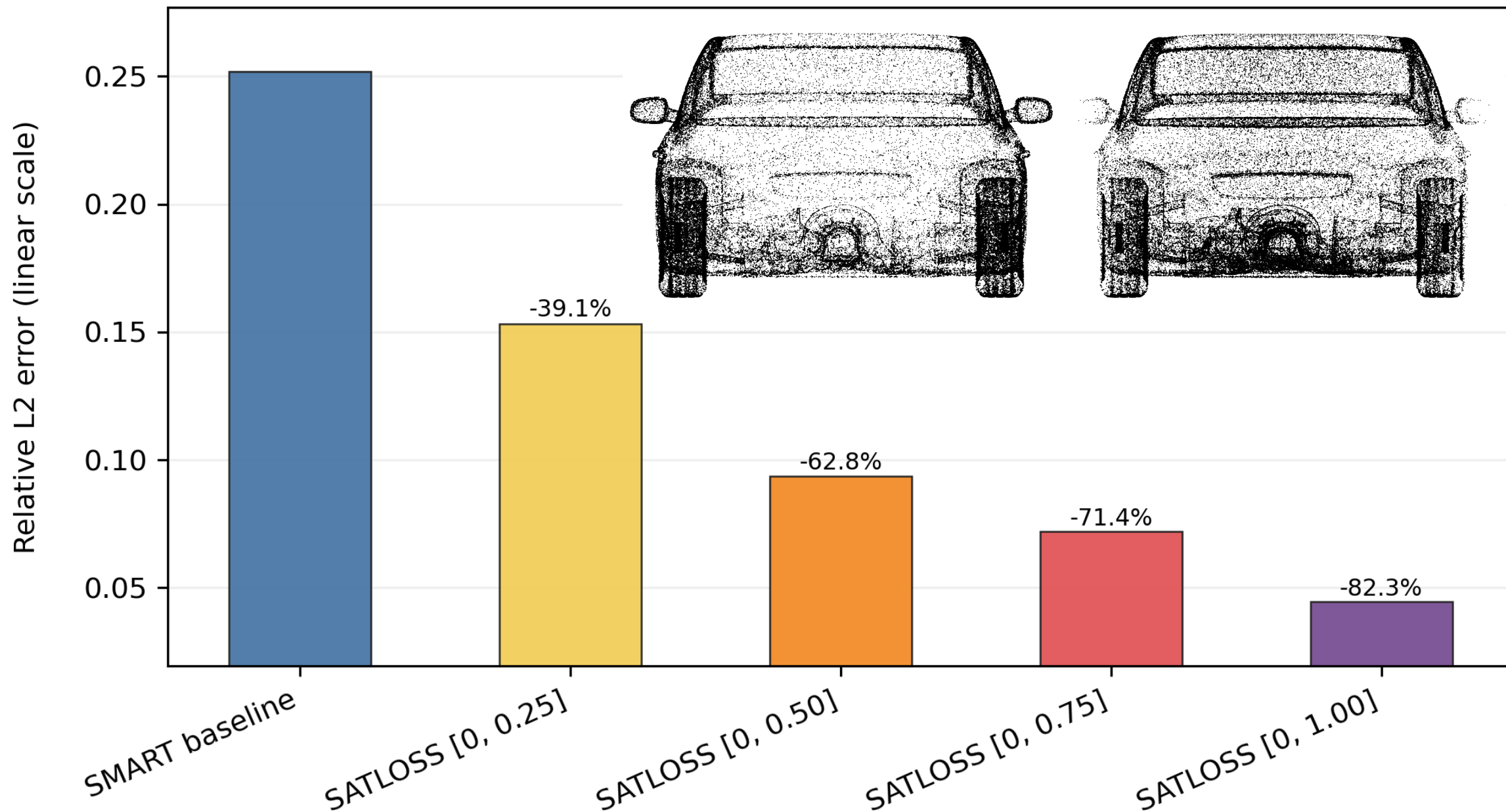
beta = 0.0 (no shift)



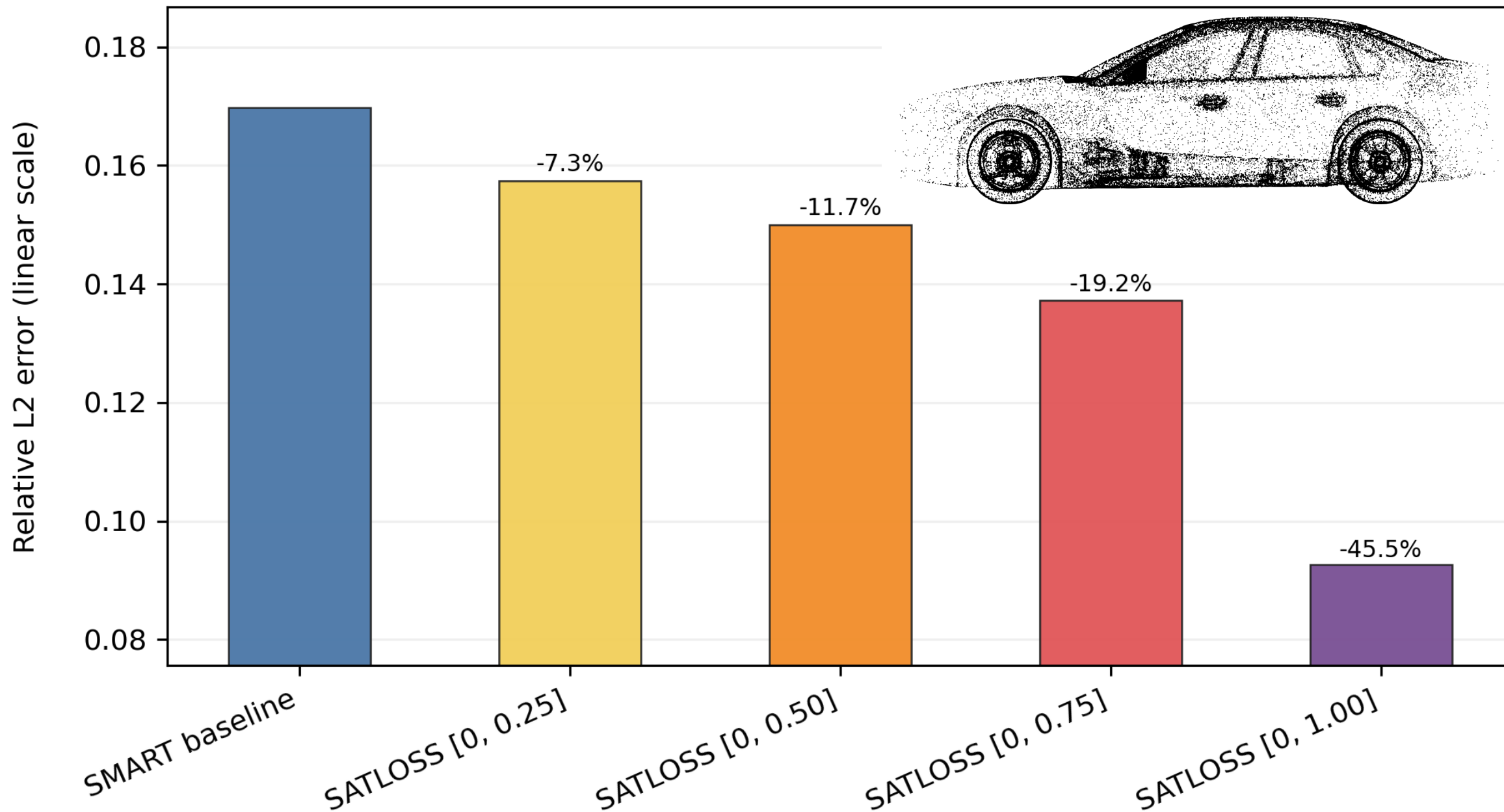
beta = 1.0

$$\text{sampling probability from full point cloud} = \frac{1}{\text{density}^{\text{beta}}}$$

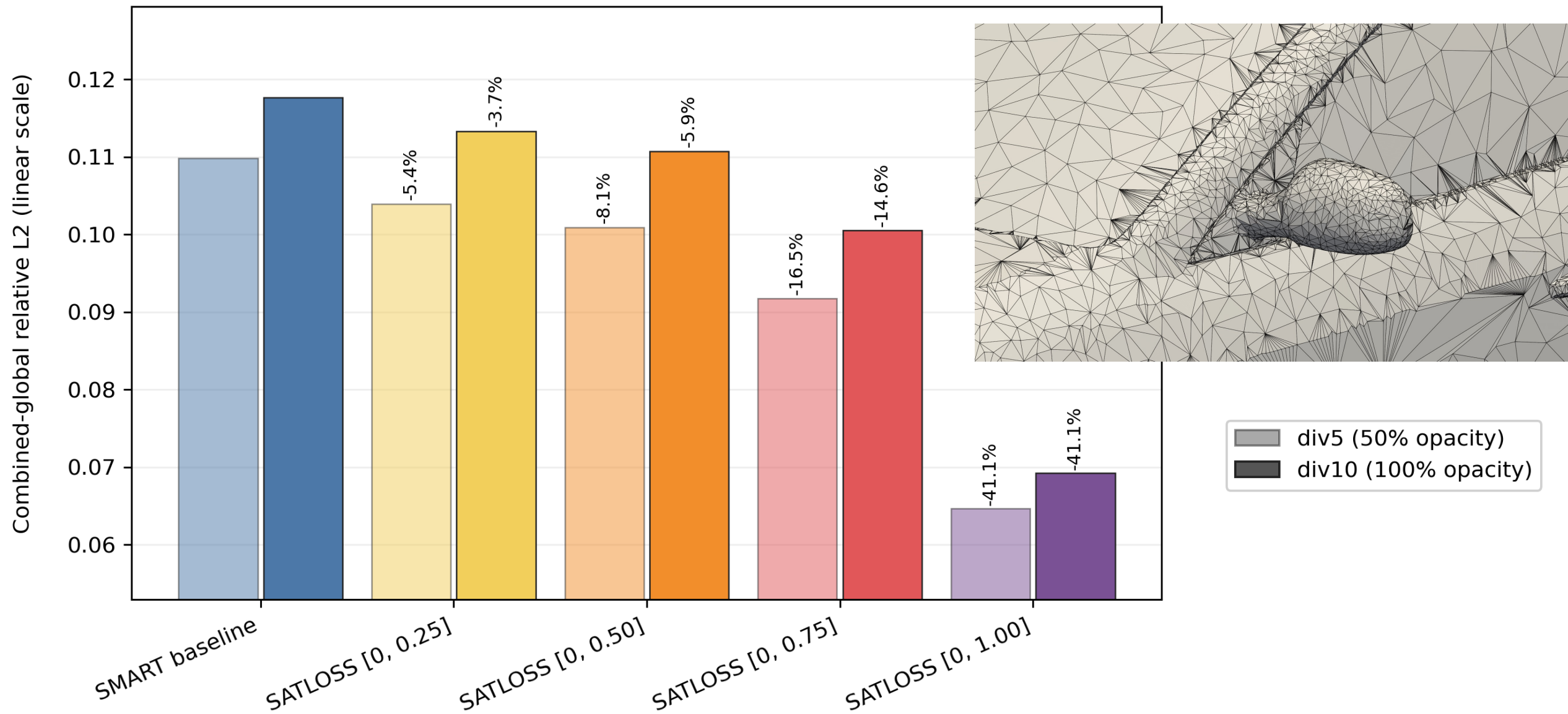
Sinusoidal-y shift at severity 1.00: combined global rel l2



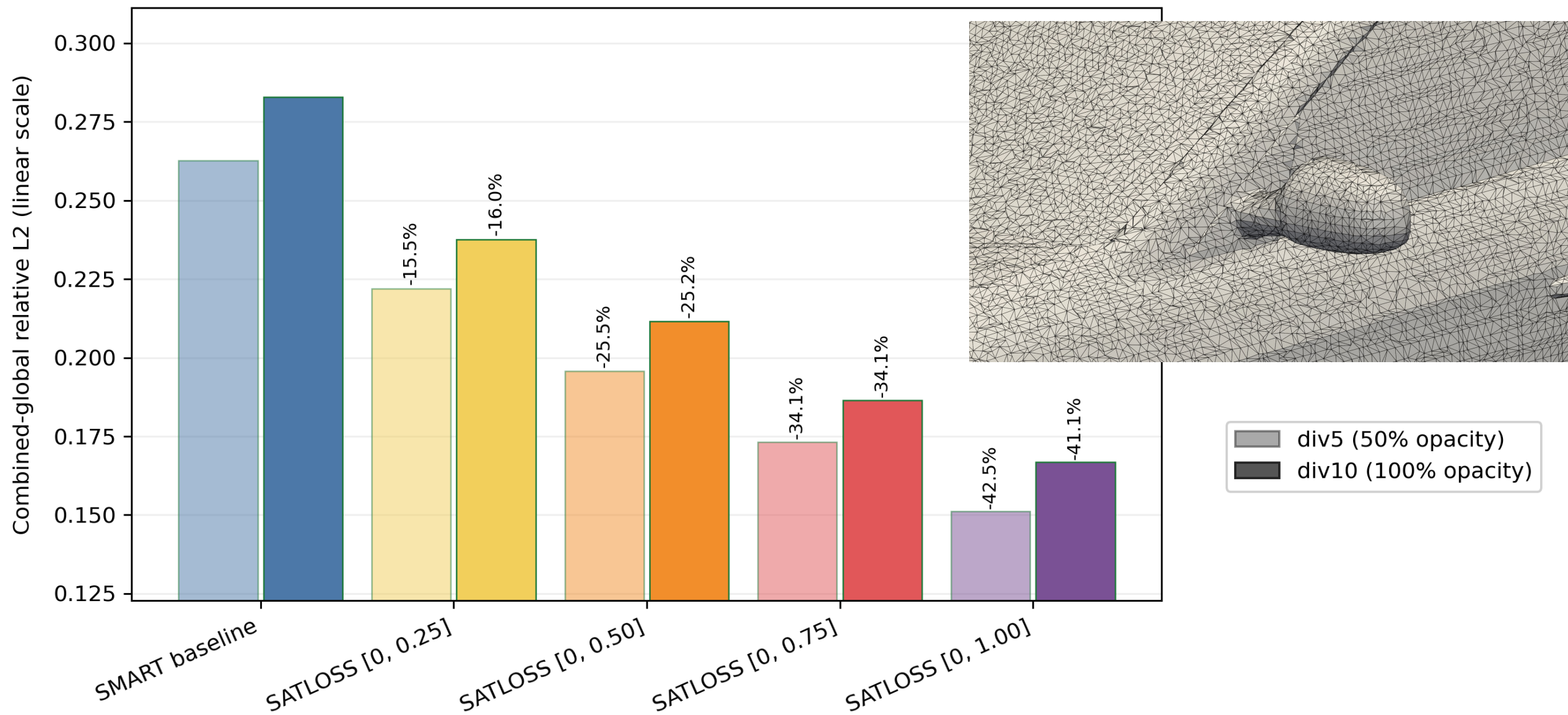
Sinusoidal-x shift at severity 1.00: combined global rel l2



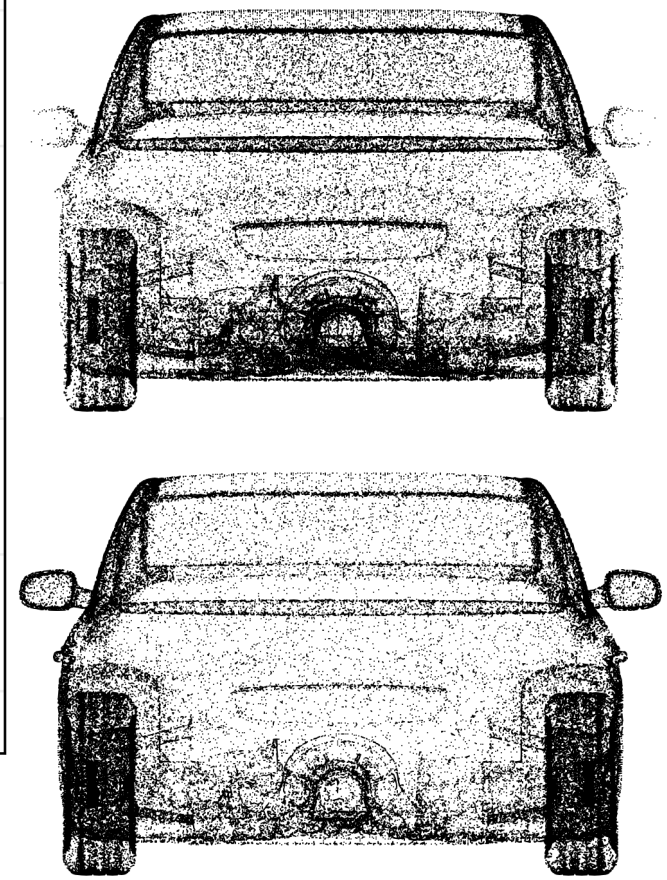
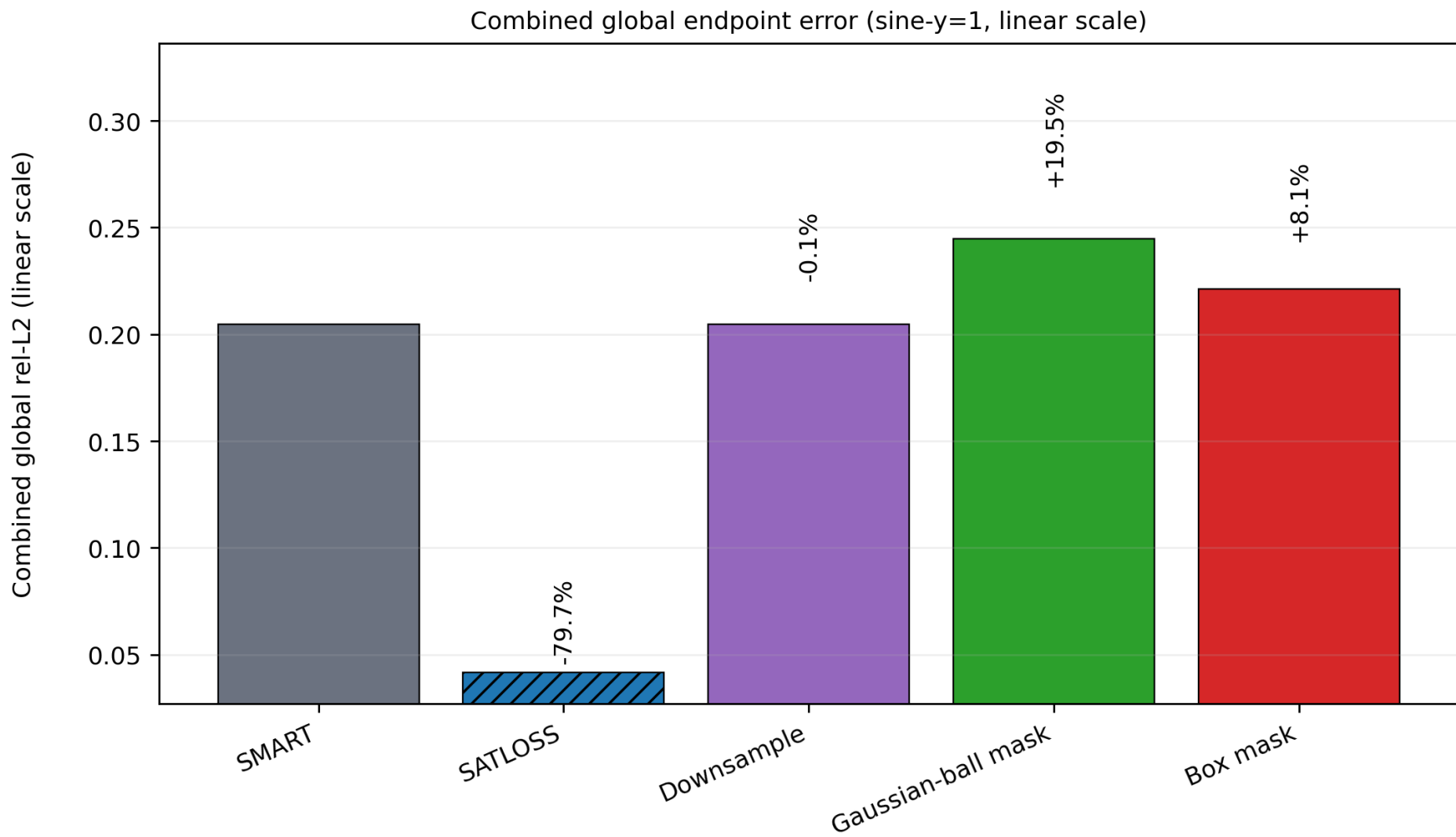
Combined global error (Angle remeshing, linear scale)

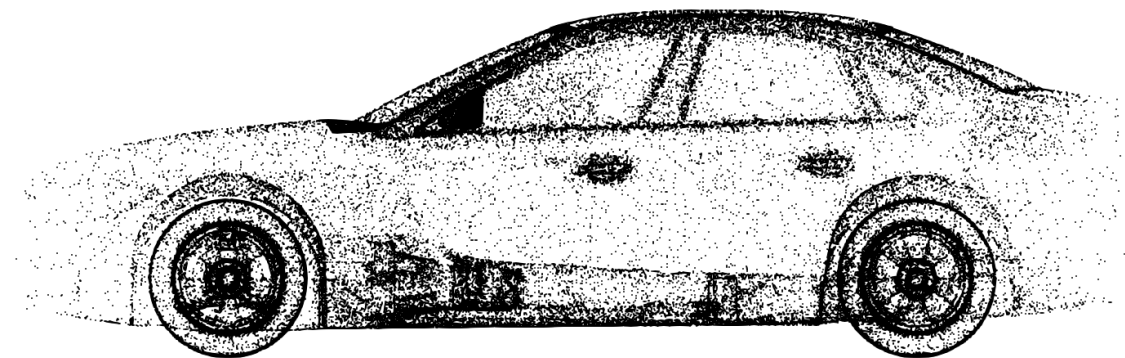
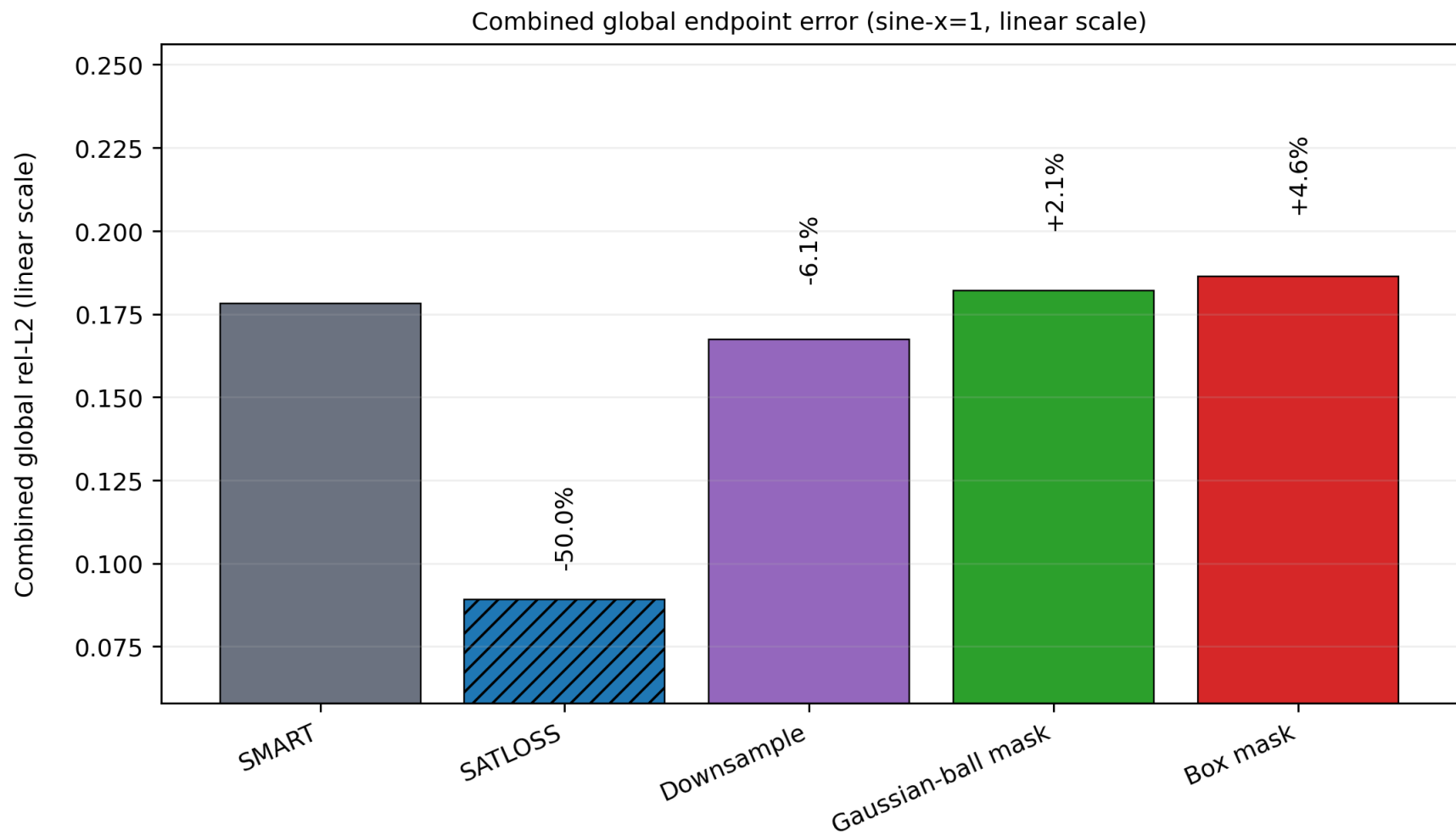


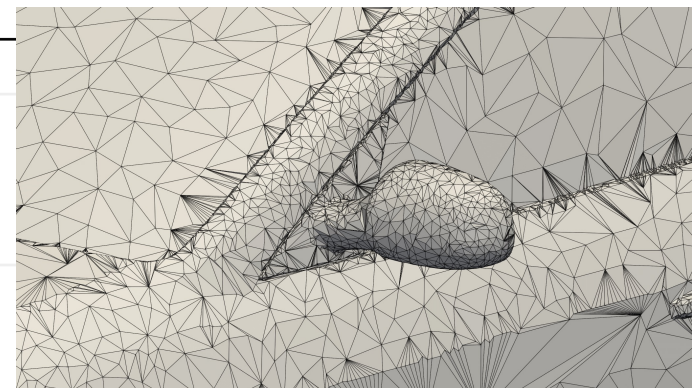
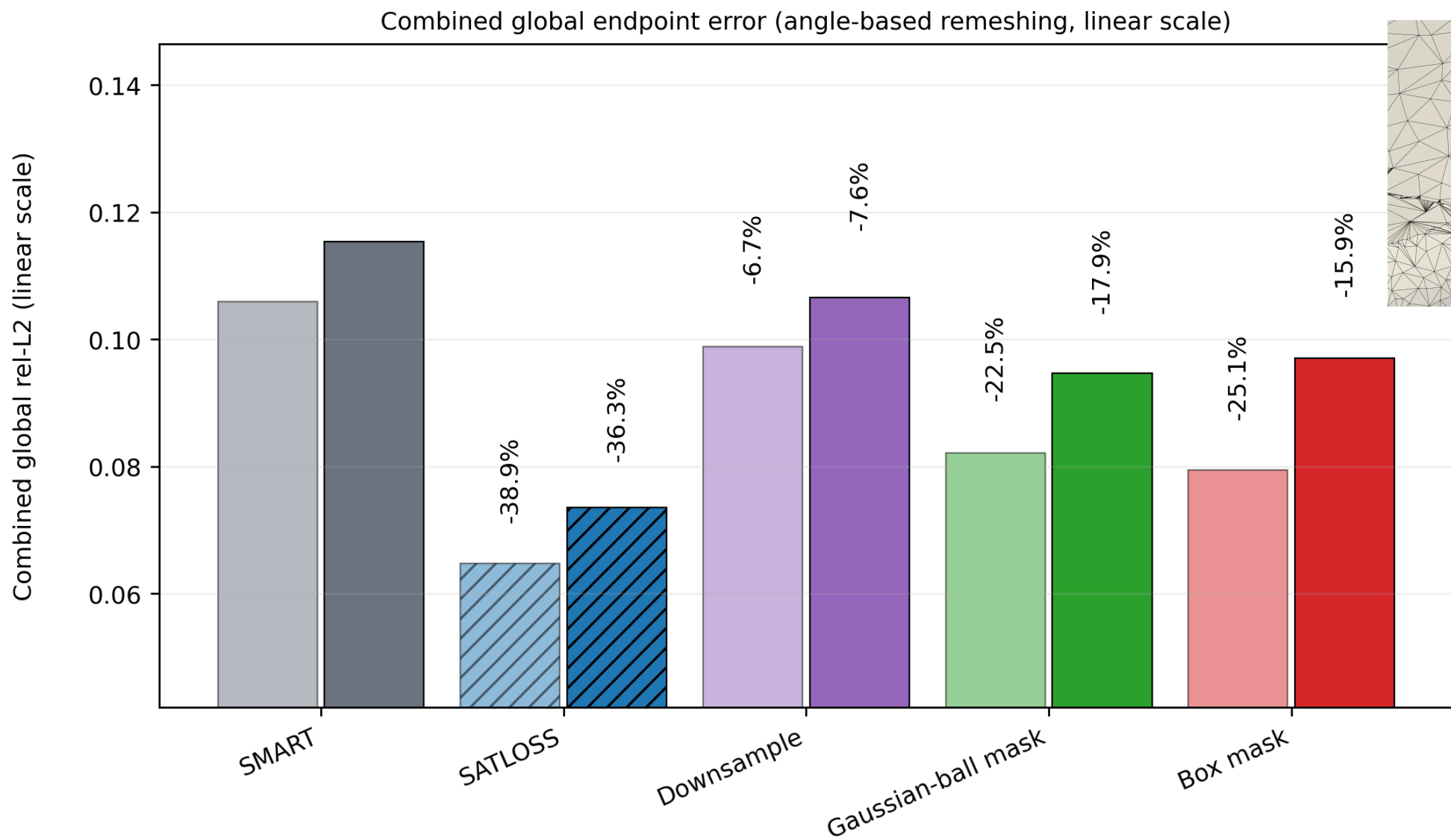
Combined global error (Isotropic remeshing, linear scale)



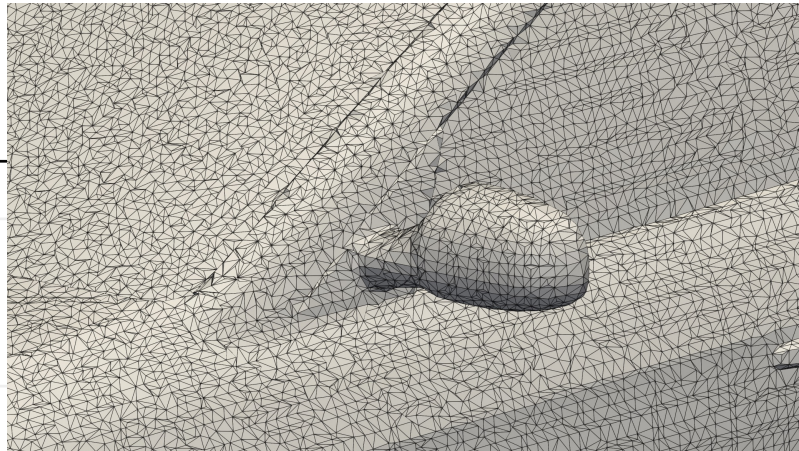
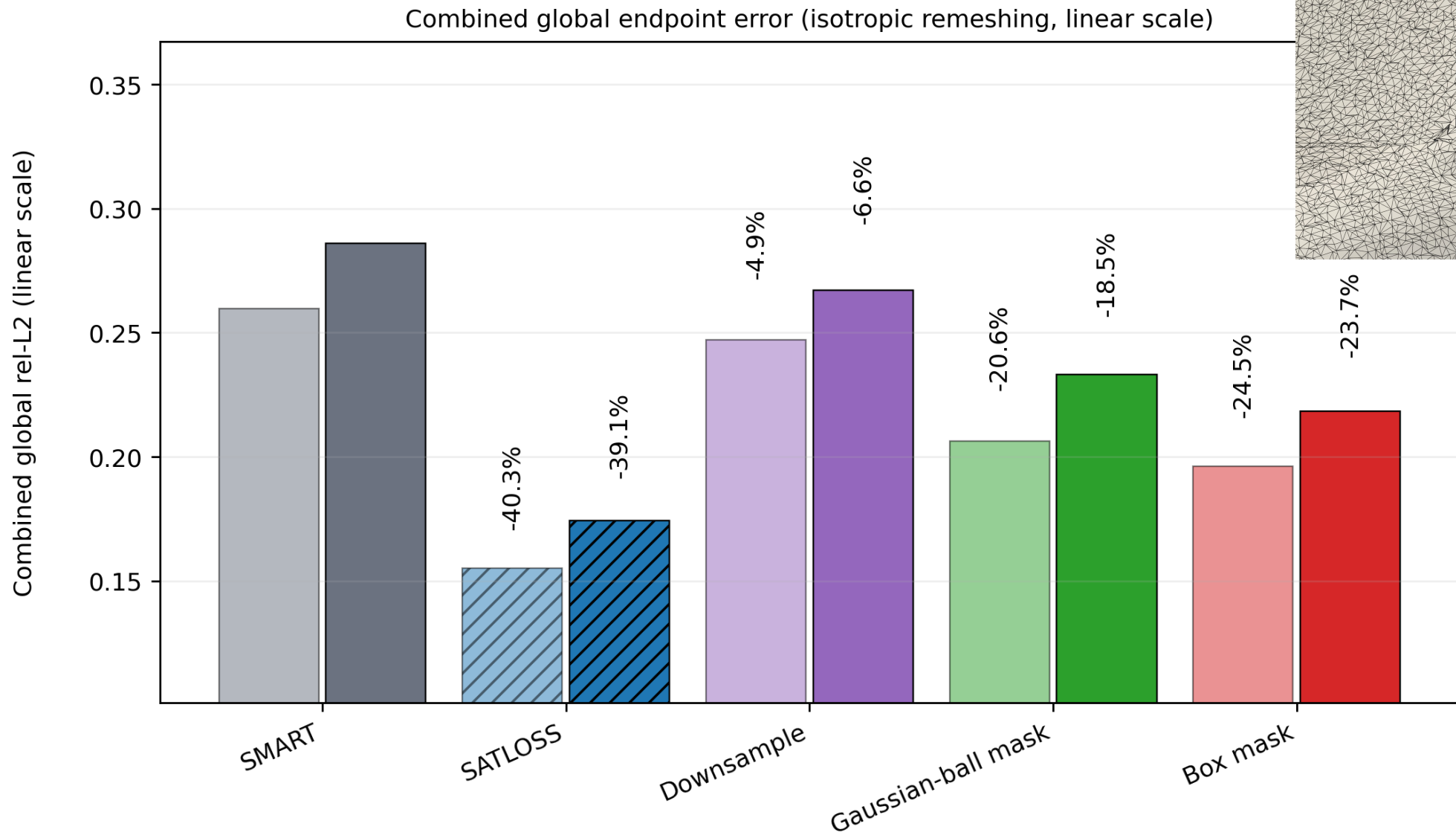
Results – vs legacy







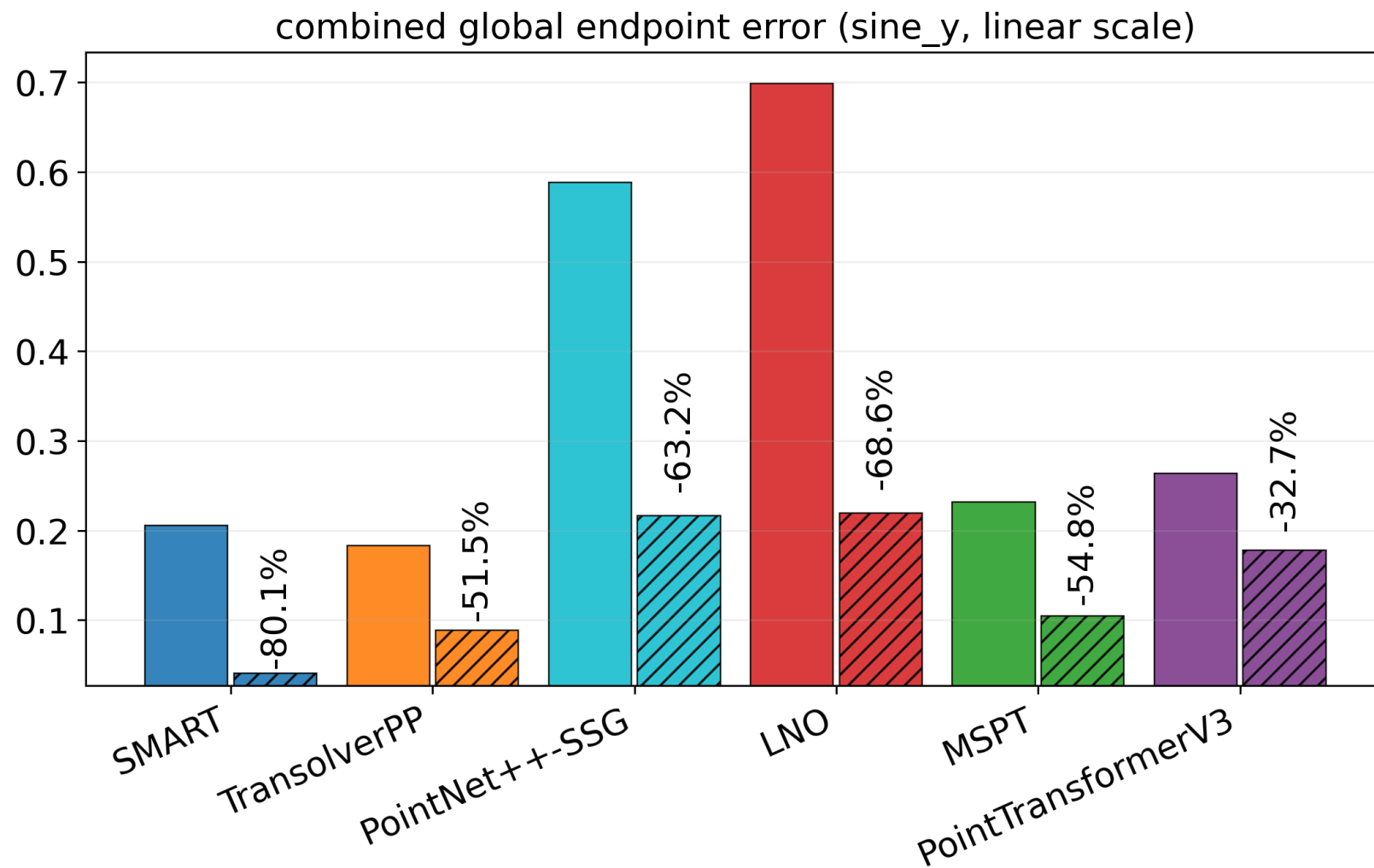
div5 (50% opacity)
div10 (100% opacity)



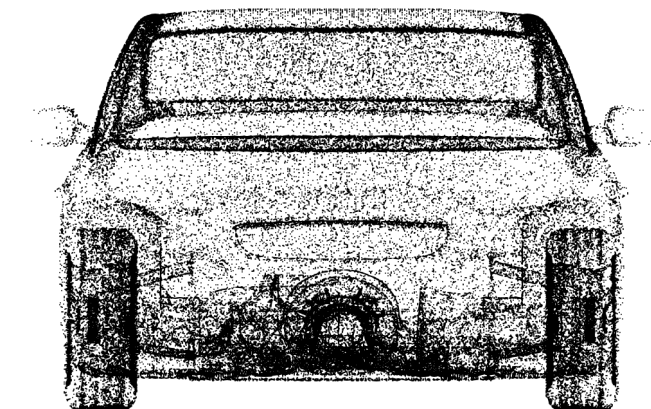
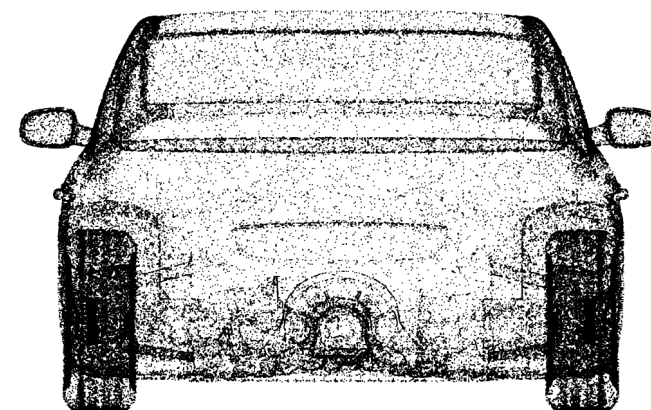
Results - Models

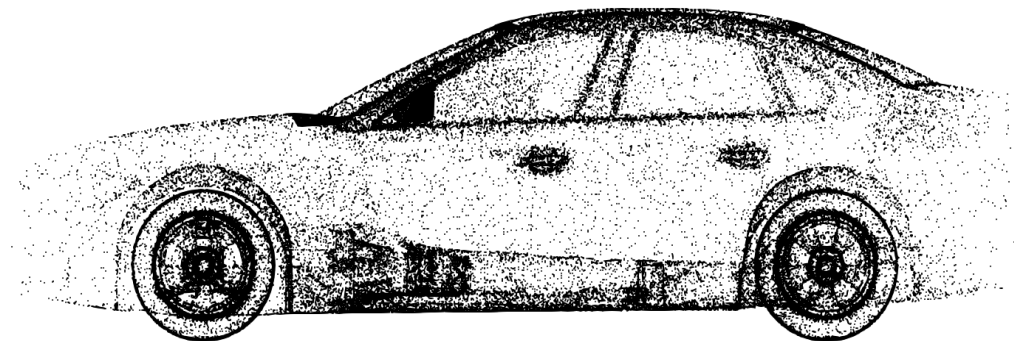
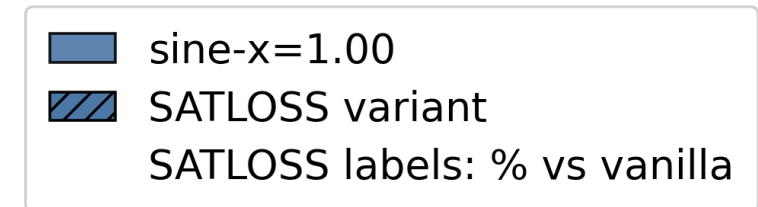
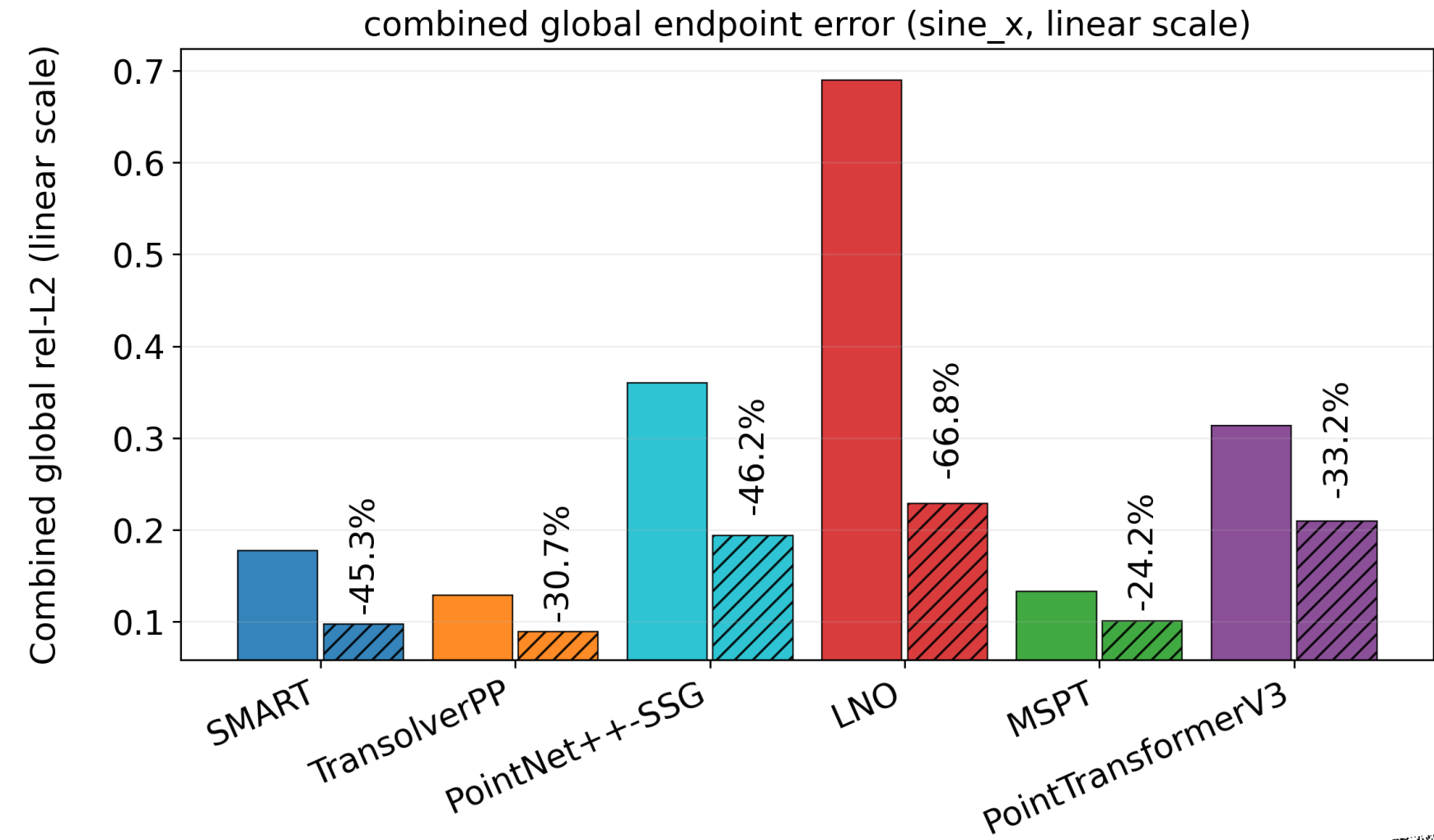
Since beta range of 0 till 1 worked the best we use that for our satloss in this section

Combined global rel-L2 (linear scale)



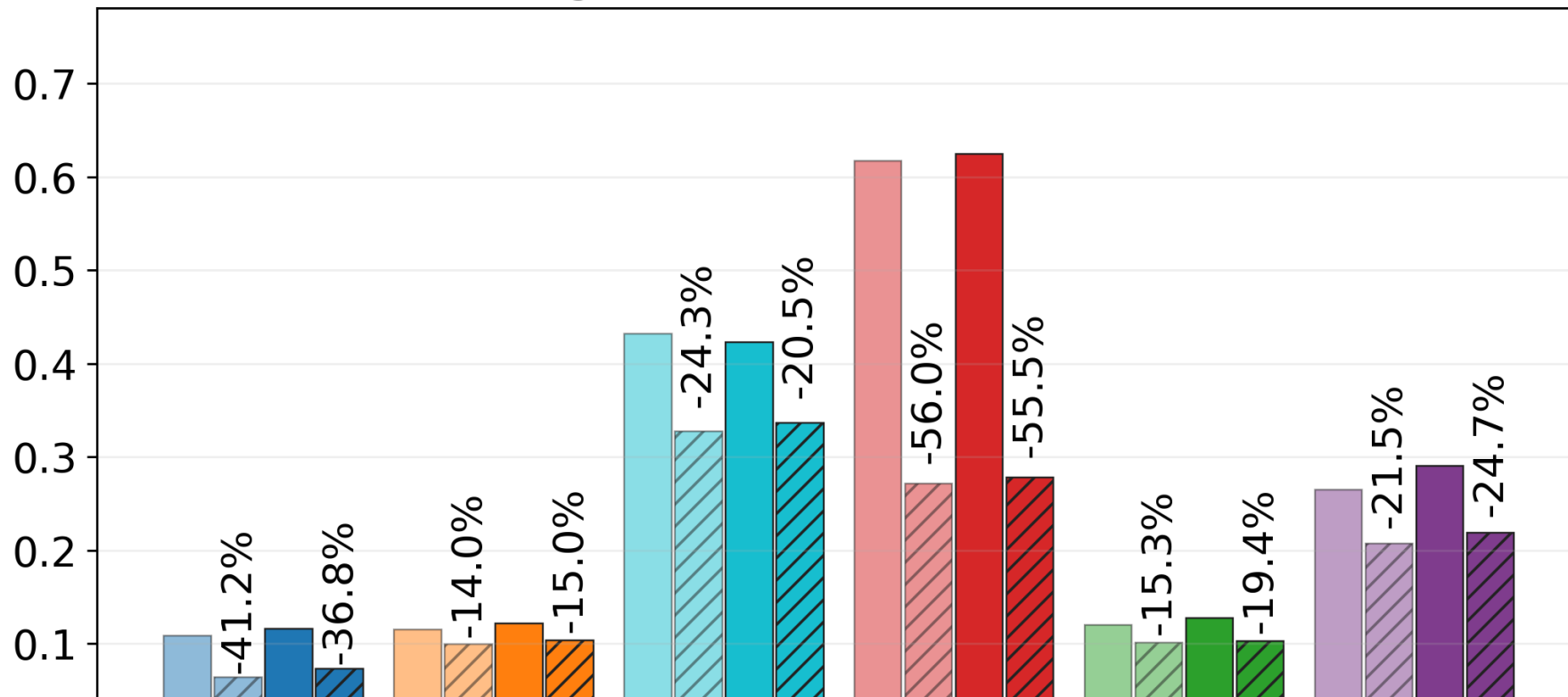
■ sine-y=1.00
▨ SATLOSS variant
SATLOSS labels: % vs vanilla





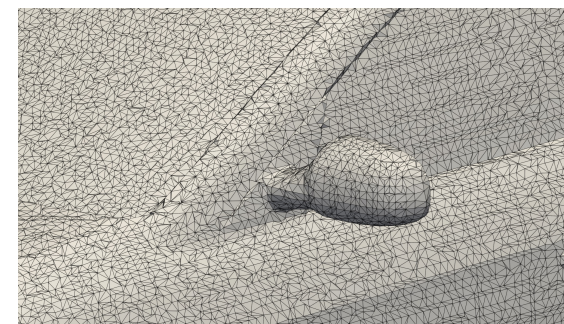
Combined global rel-L2 (linear scale)

combined global VTP-source error (linear scale)

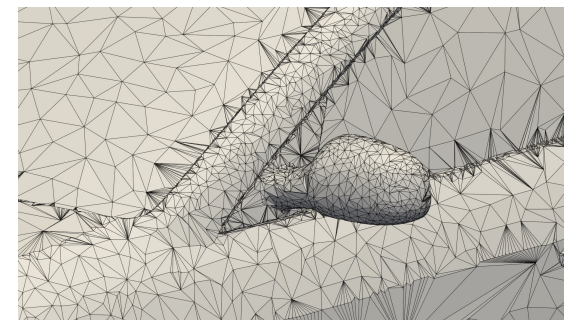


div5 50% opacity
div10 100% opacity

Vanilla (no hatch)
SATLOSS (hatch)



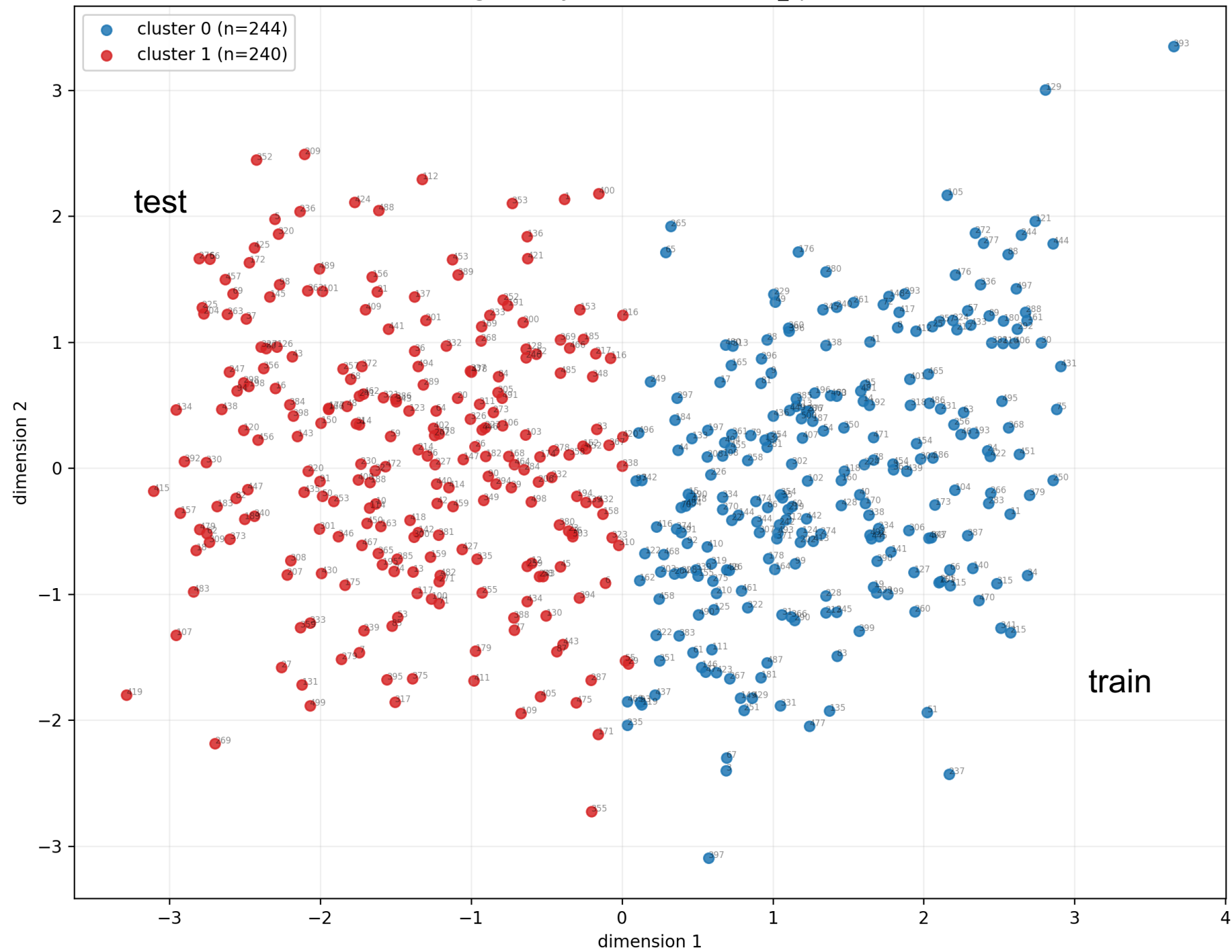
Avg Err over three methods
Isotropic, Voxel and Feature



Shift in Geometry

Let's see whether satlosss can help with shift in geometry in the same design language but further in terms of PCA

Statistical geometry clusters: multiscale_spectral (PCA)





Cluster 0 (n=244)
Cluster 1 (n=240)

